

exp: 5

16-bit Addition

AIM:

To write an assembly language program to implement 16-bit addition using 8085 processor.

ALGORITHM:

- 1) Start the program by loading a register pair with address of 1st number.
- 2) Copy the data to another register pair.
- 3) Load the second number to the first register pair.
- 4) Add the two register pair contents.
- 5) Check for carry.
- 6) Store the value of sum and carry in memory location.
- 7) Terminate the program.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 3050	Load the content of memory location 3050H into A.
MOV B, A	Move the content of the accumulator (A) into B. Now, B contains value from 3050H.
LDA 3051	Load the content of 3051H to A.
ADD B	Add the content of B to A.
STA 3052	Store the result of addition into 3052.
LDA 3053	Load content of 3052H into A.

MOV B, A

LDA 3054

ADC B

STA 3055

HLT

Move the content of A to B

Load content of memory location 3054 H to A.

Add content of B to A.

store result of addition into 3055.

Halt the execution of the program.

INPUT:

Address

3050

3051

3053

3054

data.

2

3

5

5

OUTPUT

Address

3052

3055

Data:

5

10

RESULT:

Thus, the program was executed successfully using 8085 processor simulator.

File Reset Assembler Debug Help



Registers

A	0A
BC	05 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 17
SP	FF FF
Int-Reg	00

Flag

S	0
Z	0
AC	0
P	1
C	0

Load me at

```
1 LDA 3050
2 MOV B,A
3 LDA 3051
4 ADD B
5 STA 3052
6 LDA 3053
7 MOV B,A
8 LDA 3054
9 ADC B
10 STA 3055
11 HLT
```

Decimal - Hex Conversion

Decimal Hex

0	0
→ To Hex	← To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

3054	-	+	05
Update Memory			

Data Stack KeyPad **Memory** I/O Ports

Start 3052

OK

Address (Hex)	Address	Data
0BEC	3052	5
0BED	3053	5
0BEE	3054	5
0BEF	3055	10
0BF0	3056	0
0BF1	3057	0
0BF2	3058	0
0BF3	3059	0
0BF4	3060	0
0BF5	3061	0
0BF6	3062	0
0BF7	3063	0
0BF8	3064	0
0BF9	3065	0
0BFA	3066	0
0BFB	3067	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

